

Notice vs Circular

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TITLE:      GCN/LVC NOTICE
NOTICE_DATE: Sat 08 Nov 25 12:32:18 UT
NOTICE_TYPE: LVC Preliminary
TRIGGER_NUM: S251108dz
TRIGGER_DATE: 20987 TJD; 312 DOY; 2025/11/08 (yyyy/mm/dd)
TRIGGER_TIME: 45095.942273 SOD {12:31:35.942273} UT
SEQUENCE_NUM: 1
GROUP_TYPE: 2 = Burst
SEARCH_TYPE: 1 = AllSky
PIPELINE_TYPE: 7 = MLy
FAR:      1.678e-05 [Hz] (one per 0.7 days) (one per 0.00 years)
TRIGGER_ID: 0x10
MISC:      0x1899003
SKYMAP_FITS_URL: https://gracedb.ligo.org/api/superevents/S251108dz/files/mly.multiorder.fits,0
EVENTPAGE_URL: https://gracedb.ligo.org/superevents/S251108dz/view/
COMMENTS:  LVC Preliminary Trigger Alert.
COMMENTS:  This event is an OpenAlert.
COMMENTS:  LIGO-Hanford Observatory contributed to this candidate event.
COMMENTS:  LIGO-Livingston Observatory contributed to this candidate event.
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- Machine readable
- Pre-defined format and entry keywords

The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration report:

We identified the compact binary merger candidate S251108dn during real-time processing of data from LIGO Hanford Observatory (H1), LIGO Livingston Observatory (L1), and Virgo Observatory (V1) at **2025-11-08 10:32:43.578 UTC** (GPS time: 1446633181.578). The candidate was found by the Aframe [1], cWB BBH [2], GstLAL [3], MBTA [4], PyCBC Live [5], and SPIIR [6] analysis pipelines.

S251108dn is an event of interest because its false alarm rate, as estimated by the online analysis, is $9.4\text{e-}18$ Hz, or about one in $1\text{e}10$ years. The event's properties can be found at this URL:

<https://gracedb.ligo.org/superevents/S251108dn>

The classification of the GW signal, in order of descending probability, is BBH (>99%), NSBH (<1%), Terrestrial (<1%), or BNS (<1%).

Noise transients (glitches) were present in the LIGO Livingston and Virgo detector data within two seconds of the event time, which may affect the parameters or the significance of the candidate.

Assuming the candidate is astrophysical in origin, the probability that at least one of the compact objects is consistent with a neutron star mass (HasNS) is <1%. [7] Using the masses and spins inferred from the signal, the probability of matter outside the final compact object (HasRemnant) is <1%. [7] Both HasNS and HasRemnant consider the support of several neutron star equations of state for maximum neutron star mass. The probability that either of the binary components lies between 3 and 5 solar masses (HasMassGap) is 2%.

The source chirp mass falls with highest probability in the bin (5.5, 11.0) solar masses, assuming the candidate is astrophysical in origin.

- Created by humans to be read by humans
- Contain ad-hoc info about the specific transient.
- Can be subject to correction
- Now grouped by single event. Let's give a look to [GW170817/GRB170817A](#)

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